COMPUTER ENGINEERING

QUEZON CITY

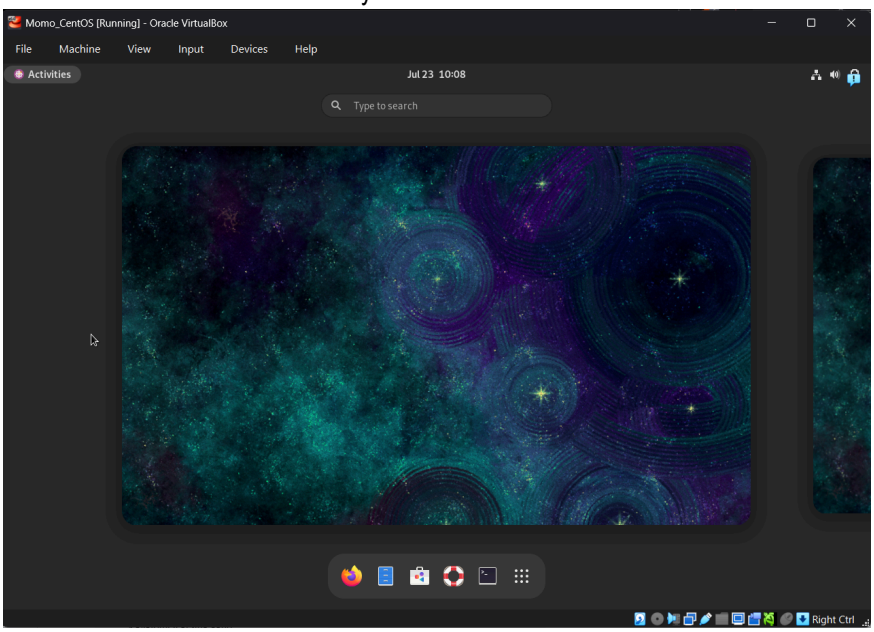
COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	7/23/2026	**DATE SUBMITTED:**	7/23/2026
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ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:
$ dnf install openssh-server

```
[root@localhost bmomo]# dnf install openssh-server
CentOS Stream 9 - BaseOS                        1.1 MB/s | 9.0 MB     00:08
CentOS Stream 9 - AppStream                     4.1 MB/s |  28 MB     00:06
CentOS Stream 9 - Extras packages                29 kB/s |  21 kB     00:00
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost bmomo]#
```

2. Start the sshd daemon and set to start after reboot:
$ systemctl start sshd
$ systemctl enable sshd

```
[root@localhost bmomo]# systemctl start sshd
[root@localhost bmomo]# systemctl enable sshd
```

3. Confirm that the sshd daemon is up and running:
$ systemctl status sshd

```
[root@localhost bmomo]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-07-23 10:06:46 PST; 6min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 995 (sshd)
      Tasks: 1 (limit: 22965)
     Memory: 1.7M (peak: 1.9M)
        CPU: 17ms
     CGroup: /system.slice/sshd.service
             └─995 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 10:06:46 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 23 10:06:46 localhost.localdomain sshd[995]: Server listening on 0.0.0.0 po>
Jul 23 10:06:46 localhost.localdomain sshd[995]: Server listening on :: port 22.
Jul 23 10:06:46 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

4. Open the SSH port 22 to allow incoming traffic:
$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

```
[root@localhost bmomo]# firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@localhost bmomo]# firewall-cmd --reload
success
[root@localhost bmomo]#
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

```
# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

- I changed the PermitRootLogin from password-prohibited to no.

$ systemctl reload sshd

```
[root@localhost bmomo]# sudo nano /etc/ssh/sshd_config
[root@localhost bmomo]# systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.
2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
momo1@workstation:~$ ssh-copy-id bmomo@192.168.56.102
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be establis
ED25519 key fingerprint is: SHA256:i2nlXTy7PCbkOeOSiLYOAz3gDN9OVSKuOoRLzXpki
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to fil
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are pr
ed now it is to install the new keys
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
bmomo@192.168.56.102's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'bmomo@192.168.56.102'"
and check to make sure that only the key(s) you wanted were added.

momo1@workstation:~$
```

3. On CentOS, verify that you have the authorized_keys.

```
[bmomo@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK7Kov7I86dHj/Jqny2k6EVGbh6jnqer3AhbIhgAQuvU
 momo1@workstation
[bmomo@localhost ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
momo1@workstation:~$ ssh bmomo@192.168.56.102
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 12:40:18 2026
[bmomo@localhost ~]$
```

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- In choosing the best distribution between Debian and Red Hat Linux, we should think of how we will use it first because each distros have their own purpose. We should also check if the distros are easy to learn since there are many distributions and some of them are kind of hard to configure. Also, Debian is free, while Red Hat usually has a paid subscription if we want full support, so we should take into consideration whether we need a distribution that has a full support but has a fee or just the free one. Overall, choosing the distributions depends on how we will use it and our preferences in terms of difficulty of managing them.

2. What are the main differences between Debian and Red Hat Linux distributions?

- As I mentioned, the big difference is that Debian is free, while Red Hat has a subscription since it is made for businesses and it has full support. Another difference is that Debian uses APT package manager with the deb packages, while Red Hat uses DNF/YUM with the .rpm packages. Additionally, Debian is known for its personal use and learning Linux while Red Hat is commonly used in companies.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Summarization**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The AI tool was only used as a guide for the summarization of articles online. It was not used to generate answers and my answers were based on my understanding.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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